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Tax Compliance as an Evolutionary Coordination Game: An Agent-Based Approach

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Abstract

Tax reporting compliance by small business owners is modeled in an agent-based framework using concepts and methods based on evolutionary dynamics. A business owner's "fitness" is a function of net after tax (and post-audit) income. Business owners exhibit heterogeneous tax morale and compliance propensity following four stochastically assigned behavioral "archetypes": Honest, Strategic, Defiant, and Random. The model is calibrated to observations from laboratory experiments and taxpayer random audits. The calibrated model is used to simulate evolutionary changes in a static population of 10,000 small business owners. A simulation using realistic parameters for the probability of audit and penalty rate finds that after fifteen time periods, the initial number of Honest business owners declines by approximately one-third and are displaced by proprietors having either Defiant or Strategic compliance behaviors.

Keywords

tax compliance, agent-based modeling, evolutionary game theory, laboratory experiments, random taxpayer audits

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Introduction

For at least two decades, it has been known that taxpayers are significantly more compliant than economic theory predicts. For example, Christian (1994) found that more than half of taxpayers randomly selected for audit in the U.S. Internal Revenue Service's (IRS) 1988 Taxpayer Compliance Measurement Program (TCMP) did not underreport their tax obligations. In contrast, economic models based on expected utility (EU) theory (Allingham and Sandmo 1972; Srinivasan 1973) predict that rational taxpayers should underreport some or all tax due,[1] given the low probability of detection (audit) prevalent in most countries.

One explanation for this anomaly is given by Erard and Feinstein (1994) who assume individuals experience guilt when they contemplate tax evasion and shame when they are caught by the authorities. Erard and Feinstein hypothesize that many people may be deterred from making the economically rational decision to evade by the emotional disutility stemming from these "moral sentiments."[2] Bernasconi (1998) doubts the deterrent effects of guilt and shame are so high as to overcome the economic incentives of tax evasion, especially since the tax agency does not typically publicize the names of noncompliant taxpayers (except in criminal cases). Instead, he finds it more likely that individuals overweight the probability of rare events (such as being selected for a tax audit) as has been observed in laboratory experiments (Tversky and Kahneman 1992; Camerer and Ho 1994; Wu and Gonzalez 1999). According to Bernasconi, taxpayers become compliant at lower audit rates than predicted by EU theory when a nonlinear weighting function is assumed for the perceived audit probability.

While laboratory experiments have shown how individuals tend to overestimate the probability of rare events, which could partly explain the higher than predicted levels of tax compliance, other research finds that individuals exhibit much heterogeneity in their tax reporting behavior (Alm, McClelland, and Schulze 1992). This finding is supported by Erard and Ho (2003) who examined micro-level TCMP data and found that taxpayers exhibit highly diverse behavior ranging from the "pathologically honest" to the "flagrantly defiant" even among persons with ample opportunity to evade, such as informal suppliers of services and goods.

The explanations thus far have focused on psychological departures from rationality to account for higher than predicted taxpayer compliance. An alternative view is proposed by Alm and McKee (2004) who find parallels between taxpayer reporting behavior and the failure of subjects in a coordination game to achieve Nash equilibrium due to strategic uncertainty. Using

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laboratory experiments, Alm and McKee show that when tax returns are selected for audit based on the deviation of each individual's reported tax from the average of all other taxpayers,[3] individuals find it difficult to coordinate on the zero compliance Nash equilibrium. The authors demonstrate that Nash equilibrium can be achieved when pregame communication among subjects is allowed.

While the results reported by Alm and McKee (2004) are intriguing, their findings are based on observations of just five subjects per experimental session. Other research has shown coordination becomes more difficult to achieve as group size increases (Van Huyck, Battalio, and Beil 1990, 1991). Thus, it remains an open question whether the coordination game analogy is an appropriate one in the tax compliance context.

An alternative theoretical framework that does not rely on direct communication among individuals is evolutionary dynamics. An article by Miller and Andreoni (1991) demonstrated that the repeated application of simple learning rules by boundedly rational agents can reproduce the convergence to Nash equilibrium observed in public goods laboratory experiments. However, as noted by Clemens and Reichmann (2006), the study by Miller and Andreoni cannot explain why perfect free riding is not observed in public goods games. Clemens and Reichmann suggest that alternative learning strategies used by subjects in laboratory experiments (and, by extension, in the real world) could account for the persistent deviation from the Nash solution. In particular, the authors propose that individuals use a strategy of *learning by experiment*, which leads to a rapid convergence toward zero public goods contributions but also never comes to a rest, as individuals strive to achieve marginal improvements in utility. An extension of the approach by Clemens and Reichmann would be to allow for agents with heterogeneous behaviors.

This article describes and implements an agent-based model of tax reporting compliance using concepts and methods of evolutionary dynamics. An advantage of agent-based models over laboratory experiments is the ability to collect data from large populations of heterogeneous agents. In the model, each "agent" is an owner of a small business (Schedule C proprietor), whose behavior conforms to one of four distinct reporting compliance "archetypes" (Stahl and Wilson 1995). These are: Honest, Defiant, Random, and Strategic. Honest and Defiant archetypes play a fixed strategy while the Random archetype plays a random strategy with no learning behavior. The Strategic archetype uses a simple reinforcement learning algorithm to determine how much income to report to tax authorities. A firm's fitness for survival is positively correlated to net after tax (and post-audit) income with a probabilistic selection rule. In addition,

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the model assumes some firms cease operating for noneconomic reasons (e.g., retirement and voluntary career change).

The model is calibrated to observed behavior in laboratory experiments. The calibrated model is then used to simulate the compliance characteristics of a population of 10,000 small business owners. Output from different simulations using alternative assumptions for risk aversion and neighborhood effects (i.e., knowledge of the tax compliance behavior of those in one's immediate social network) are compared to observed tax compliance characteristics of actual small business owner/operators. Using real-world values for the probability of audit (p) and fine (penalty) rate (ϕ), the simulation results indicate that small businesses exhibit risk aversion but do not appear to condition tax compliance reporting decisions on their neighbors' behavior (Fortin, Lacroix, and Villeval 2007; Korobow, Johnson, and Axtell 2007; Zaklan, Westerhoff, and Stauffer 2009).

The next section briefly reviews the economic model of tax compliance and compares the model's theoretical predictions to observed behavior from taxpayer random audit studies and laboratory experiments. The third section describes the major features of an agent-based model of tax reporting compliance for small business taxpayers (the Small Business Tax Compliance Simulator—SBTCS). The fourth section contains a discussion of model calibration and an example that uses the calibrated model to explore long-run compliance behavior of a population of 10,000 small business owners. The last section concludes.

Tax Compliance Behavior: Theory and Evidence

In the economic model of tax compliance (Allingham and Sandmo 1972; Srinivasan 1973; Yitzhaki 1973), a taxpayer must decide how much income to report to the tax authority to maximize EU. In mathematical form,[4] a taxpayer's EU from reporting x dollars of income is:

$$\mathrm{EU} = (1-p)(y-tx)^{\alpha} + p[y - ty - \phi(ty - tx)]^{\alpha}. \quad (1)$$

Where p is the probability of detection (usually equated with the "audit" rate), t is the marginal tax rate, y is annual total taxable income, ϕ is the penalty per dollar of tax underreported, and α is the taxpayer's coefficient of relative risk aversion. If not audited $(1-p)$, she enjoys her true income y less the tax on the amount of income reported tx. If audited, the taxpayer must remit the full tax liability (ty) along with a fine on the unreported amount of tax ($ty - tx$).

In the case of a risk neutral taxpayer ($\alpha = 1$), we differentiate equation (1) with respect to x and solve for values of p that yield an interior solution

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$$\frac{\partial EU}{\partial x} = (1-p)t - pt\phi > 0 \tag{2}$$

$$p + p\phi < 1 \tag{3}$$

$$p < \frac{1}{1+\phi}. \tag{4}$$

Expression (4) says that a risk-neutral taxpayer should report zero income up to the point where the probability of audit is just less than the ratio $(1/(1+\phi))$ and report 100 percent of income when p equals or exceeds this ratio. In the United States, the maximum fine rate rarely exceeds 0.50 meaning that the probability of audit would have to be at least 0.67 to induce taxpayers to fully comply. The current overall audit coverage rate (number of tax returns audited divided by the number of tax returns filed) in the United States is about 0.01. These conditions would seem to encourage taxpayers to play the audit lottery, report zero taxable income every year, and enjoy increased income in most years while only rarely experiencing a loss due to a tax audit. However, this is not the case.

Citing data from previous IRS TCMP random audit studies, Bloomquist, Plumley, and Toder (2005) report taxpayers voluntarily paid between 80 and 85 percent of true tax liability throughout the 1970s and 1980s. The latest IRS random audit study for tax year (TY) 2001 found no significant change in this behavior (IRS 2007). According to Slemrod (2007), the presence of tax withholding and third-party reporting of income information by employers and financial institutions accounts for much of the disparity in voluntary compliance between the theoretical model and observed behavior. The IRS is able to use computerized information documents to electronically verify between 80 and 85 percent of income received by taxpayers, providing the equivalent of a 100 percent audit rate on income from sources such as wages, interest, dividends, and government transfer payments (Bloomquist 2003b). Reporting compliance on the remaining 15–20 percent of income from business and investment sources is much lower than that for "matchable" income (IRS 2007) but still higher and qualitatively different than predicted by rational choice theory.

Compliance Behavior of Small Business Filers

To illustrate this last point, let us examine the compliance reporting behavior of a group of U.S. taxpayers whose only source of income is from a

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small business. The data source is the IRS' National Research Program (NRP) study of individual taxpayers for TY 2001 (Bennett 2005). The total NRP sample has 44,768 audit cases representing 125,790,958 taxpayers who filed timely tax returns for TY 2001. A subsample of this data was selected consisting of taxpayers whose income is derived solely from a Schedule C sole proprietorship (both pre and post-audit). This subsample of the NRP has 1,673 audit cases representing 1,101,977 taxpayers. This sample was further limited to those Schedule C filers with positive taxable income as determined by the examiner. This added restriction facilitates comparison with reporting behavior by subjects in laboratory experiments whose compliance decision is based on reporting none, some or all of a positive amount. This final subsample has 1,101 cases representing 559,555 individual Schedule C filers. Within this data set, there were twenty nine cases where reported taxable income exceeded the amount of taxable income determined by the examiner. These cases (representing 13,131 taxpayers) were assumed to have 100 percent reporting compliance.

Table 1 summarizes the unweighted and weighted compliance statistics for this group of taxpayers. The cell entries in the two rightmost columns of table 1 refer to the mean of the individual reporting compliance rates and the overall mean reporting compliance rate defined as the amount of reported taxable income divided by the amount of taxable income per exam. The range of taxable income per exam for this sample spans five orders of magnitude from less than US$40 to more than US$4 million. The probability of audit for all individual taxpayers in calendar year 2002 was 0.57 percent and 1.72 percent for all Schedule C filers (IRS 2002).

Figure 1 displays histograms of reporting compliance rates (unweighted and weighted) for the selected group of TY 2001 Schedule C filers. Visual inspection of both plots reveals the appearance of a bimodal distribution with an apparently random distribution of observations between the two modes (at 0 and 100 percent compliance). The presence of a small group of fully compliant individuals, even though the rational decision from a purely economic standpoint is to report zero taxable income, supports an earlier observation by Erard and Ho (2003), using TY 1988 TCMP data, who identify a similar group of apparently "pathologically honest" individuals.

Comparison with Tax Compliance Laboratory Experiments

The compliance behavior of subjects in laboratory experiments also diverges from theoretical predictions. However, the degree of similarity

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Table 1. Summary Statistics for Selected Schedule C Filers (Monetary Values in U.S. Dollars)

	Taxable Income as Reported				Taxable Income per Exam			Mean Reporting Compliance Rate	
	N	Mean	Standard Deviation	Sum ($M)	Mean	Standard Deviation	Sum ($M)	Mean of Individuals	Overall Mean
Unweighted	1,101	$5,461	$12,081	$6.1	$25,277	$132,064	$27.8	0.313	0.216
Weighted	559,555	$3,708	$9,854	$2,075.0	$16,054	$78,165	$8,983.3	0.242	0.231

Source: See text.

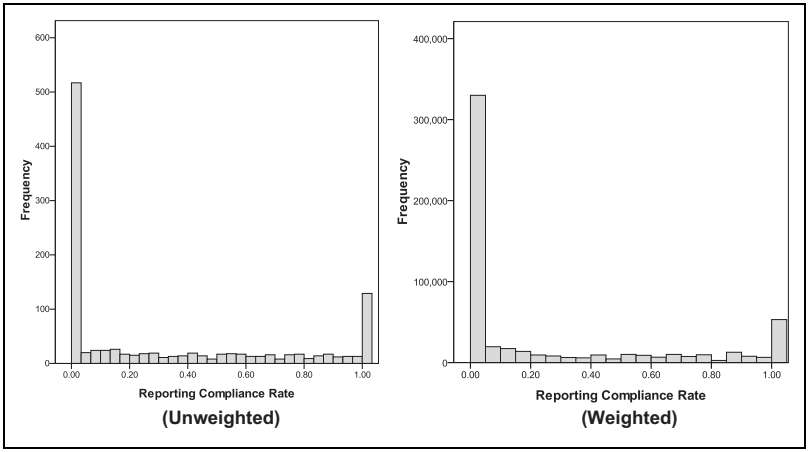


Figure 1. Distribution of reporting compliance rate for selected schedule C filers.

between the compliance behavior of small business taxpayers and laboratory subjects is remarkably similar in several ways. To illustrate this, we rely on data collected from a series of tax compliance lab experiments reported elsewhere (Alm and McKee 2006; Alm, Deskins, and McKee 2009). The full data set from these experiments has 16,560 observations from 1,072 individual experimental subjects. The experimental sessions include both "base case" or "no treatment" scenarios and several treatment scenarios (e.g., the existence of a public good, unofficial communication among participants, and "official" communication from the tax authority). For this article, data from the "no treatment" scenarios were selected. This data set contains 3,780 observations from 252 individuals. The income earned by the volunteer lab subjects in each round ranged from a low of 20 to a high of 100 lab dollars. The number of tax audits performed in each round was public information. Descriptive statistics for the full data set and the selected data set are shown in table 2.

Table 2 displays summary statistics for the five different audit probabilities used in the experiments. One session with sixteen subjects was conducted with zero probability of audit. Subjects in all sessions, including the session with zero audit probability, were informed of the audit probability prior to filing and all experiments were conducted using the same software. Figure 2A and B displays histograms of reporting compliance rates from the experimental data corresponding to the audit probabilities in table 2 (audit probability 0.40 is omitted for brevity).

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Table 2. Summary Statistics for Experiment Sample

	Full Sample			Selected Sample			
Audit Probability	*N* Subjects	*N* Observations	Mean Reporting Compliance Rate	*N* Subjects	*N* Observations	Mean Reporting Compliance Rate	Overall Mean Reporting Compliance Rate
.00	16	240	0.288	16	240	0.288	0.286
.05	180	2,700	0.413	48	720	0.404	0.368
.10	356	5,580	0.544	78	1,170	0.475	0.476
.30	298	4,710	0.590	32	480	0.558	0.536
.40	222	3,330	0.638	78	1,170	0.672	0.668
Total	1,072	16,560	0.551	252	3,780	0.521	0.517

Source: See text.

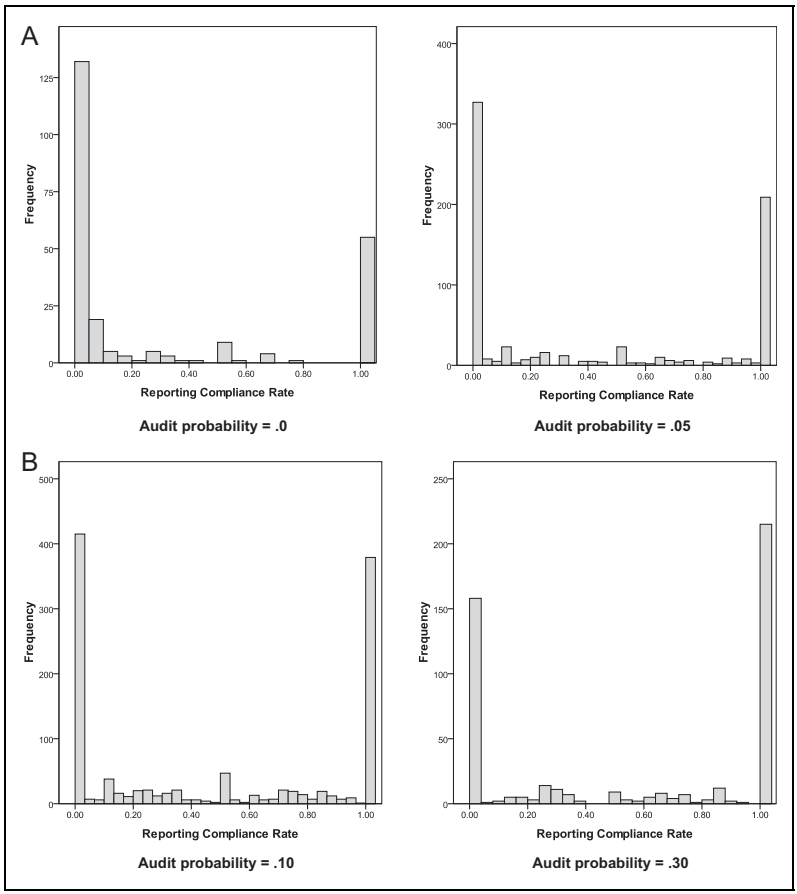


Figure 2. A, B, Distribution of reporting compliance rates for lab subjects.

These graphs exhibit the same bimodal distribution shape as the NRP sample in figure 1.

Summary

What have we learned about taxpayer compliance behavior by analyzing the compliance behavior of volunteer subjects in laboratory experiments and random audit studies? First, both the NRP and the experimental data show a small group of individuals exhibiting behavior that appears to be

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inherently "honest" in the sense that all taxable income is voluntarily reported. For example, using the data shown in table 2 for the experiment with an audit probability of 0.10, ten of seventy eight subjects (13 percent) reported 100 percent of income in each of fifteen rounds and fourteen subjects had an overall compliance rate of 95 percent or higher in all fifteen rounds. Similarly, 129 of 1,101 taxpayers (12 percent) in the subsample of NRP data were found to be fully compliant and 152 (14 percent) had a compliance rate of 95 percent or higher. However, some experimental subjects appear to be "defiant," or perhaps we should say "rational," by reporting effectively zero taxable income. Again, using data from the session with $p = .10$, there were twelve of seventy eight subjects (15 percent) whose mean compliance rate was less than 0.05 over fifteen rounds.

A second feature of the experimental data is the appearance of "all or nothing" compliance behavior. The sudden transition from zero to full compliance is predicted by EU theory for risk neutral taxpayers when $p = 1/(1 + \phi)$. However, figure 2A and B shows individuals becoming fully compliant at much lower values of p, which suggests people use different "rules of thumb" to decide when to comply. Such heterogeneous behavior is characteristic of boundedly rational individuals, in contrast to the perfectly rational decision maker in EU theory. The decision of some subjects in lab experiments to switch to full compliance at relatively low audit rates also would seem to be evidence that individuals overweight the true probability of audit (Bernasconi 1998). This hypothesis can be tested by simulating taxpayer reporting behavior with and without overweighting of audit probabilities to see which hypothesis best fits the data.

Third, we observe from the histograms of compliance rates that some individuals appear to use something other than an "all or nothing" reporting strategy; in other words, they report somewhere between 0 and 100 percent of taxable income. Again, using data from the experimental session when $p = .10$, about 68 percent of subjects had a mean reporting rate between 0.05 and 0.95 averaged over the fifteen rounds of play. However, it is not apparent whether these individuals are simply uncertain how much to report due to the complexity of the problem (e.g., uncertainty about reporting rules), risk aversion, or whether they are engaged in strategic behavior. The fact that subjects in tax compliance lab experiments face a much simpler tax scenario than actual taxpayers and exhibit qualitatively similar behavior as real taxpayers suggests that perhaps tax code complexity is not the sole explanation. Alm (1999) observes that the same individual in a tax compliance experiment can display a wide range of behavior that appears at times rational, sometimes strategic and other times seemingly unrelated to

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Table 3. Number of Lab Subjects Reporting Zero Dollars in the First and Last Rounds

	Reporting Zero Dollars of Taxable Income			
Audit Rate	Sample Size	Round 1	Round 15	Percentage Change
0.00	16	4	7	75.0
0.05	48	5	23	360.0
0.10	78	19	26	36.8
0.30	32	5	14	180.0
0.40	78	14	19	35.7

Source: See text.

external influences. Therefore, it seems reasonable to conclude that some individuals reporting between 5 and 95 percent of income are engaging in a combination of random and strategic behaviors.

Finally, it is plausible to infer from the experimental data that learning is taking place. Although participants in these experiments were not allowed to communicate with each other and were not able to observe their neighbors reporting behavior, more laboratory subjects reported zero dollars in round fifteen of the experiments than in round one. These observations are summarized in table 3 below. This pattern of behavior suggests subjects learn through a process of repeated play that the chances of being audited are not as great as they initially feared. Over time they become more confident in their ability to underreport and not get caught. Such observations are only possible in a laboratory setting and illustrate the value of using a variety of analytical tools—including lab experiments, field studies, and econometric methods—to study taxpayer behavior.

Model Description

This section describes an agent-based model (referred to as the SBTCS) that simulates the behavior of a population of 10,000 taxpayers whose only source of income is derived from a small business. The model assumes taxpayers belong to one of four reporting compliance archetypes: Defiant, Honest, Random, and Strategic. SBTCS allows users to specify alternative return selection strategies for conducting audits and has several additional options to aid in model calibration such as (1) risk aversion, (2) neighborhood effects, and (3) alternative income distributions (a uniform

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distribution used in laboratory experiments and a lognormal distribution for the natural occurring world). Each of these features is described below.

Taxpayer Behavior Archetypes and Reporting Compliance Behavior

Based on the observations of individuals' reporting behavior from taxpayer random audit studies and tax compliance laboratory experiments discussed above, taxpayer agents belong to one of four compliance archetypes (A_i): Defiant, Honest, Random, and Strategic. The variable z denotes the fraction of taxable income reported by agents of type i. For Honest agents, z is always unity. Defiant agents always report zero income ($z = 0$) unless the audit rate (p) is greater than or equal to the ratio $p = 1/(1 + \phi)$ in which case $z = 1$. In each time period, taxpayer agents of the Random archetype draw z from a triangular distribution with a mode of one and lower and upper bounds of zero and one, respectively. This distribution assumes misreporting by taxpayers is related to tax code complexity that results in a falling frequency of errors away from the full compliance level.

Taxpayer agents with the Strategic archetype initially report a random percentage of income $z_1 \sim U(0, 1)$. Stahl and Wilson (1995) refer to this as level 0 best response behavior when individuals are uncertain about how other participants in a coordination game (especially coordination games with no prior communication) are likely to behave. If not audited in time t, then in time $t + 1$, Strategic agents report $z_{t+1} = z_t \times U(0, 1)$. If audited in time t, then in time $t + 1$, Strategic agents report $z_{t+1} = z_t + U(0, 1)$ with z restricted to values ≤ 1. This simple reinforcement learning algorithm assumes a Strategic agent derives greater utility from paying less tax but an audit temporarily causes the agent to become less "aggressive." However, this model assumption may not necessarily portray how all taxpayers behave following an audit. For example, several researchers have noted the presence of a "bomb crater" effect whereby taxpayers audited in time period t actually increase the level of evasion in time period $t + 1$ (Mittone 2006; Kastlunger et al. 2009). This behavior is not represented in the current model but could be added by randomizing the response of Strategic taxpayers following an audit.

In addition to the above default compliance reporting behaviors, Random and Strategic taxpayers are assumed to exhibit "all or nothing" compliance when the real or perceived audit rate exceeds some randomly assigned threshold value, r. For example, when $p = .05$, it is assumed 5 percent of Random and Strategic taxpayers become fully compliant.

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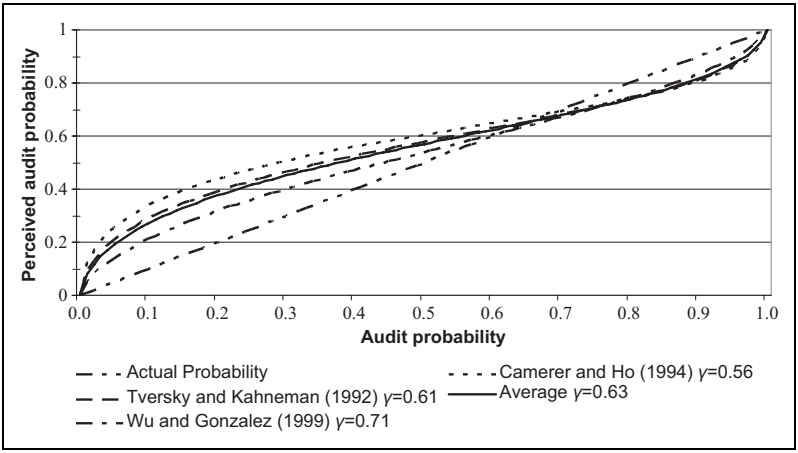


Figure 3. Weighting function for audit risk aversion.

All taxpayers, including Defiant taxpayers, become fully compliant when $p = 1/(1 + \phi)$. In other words, all agents ultimately are rational.

Risk Aversion

The SBTCS model portrays taxpayer risk aversion as overweighting the probability of a tax audit. This is modeled using equation (5) that computes the taxpayer's perceived audit probability ($\hat{p}$) by reweighting the actual audit probability using a single parameter formula:

$$\hat{p} = 1 - \left[(1-p)^{\gamma}/(p^{\gamma} + (1-p)^{\gamma})^{\frac{1}{\gamma}}\right]. \quad (5)$$

A plot of equation (5) for alternative parameter values is displayed in figure 3. The model uses a default value of 0.63 for γ, which is an average (shown as the solid line in figure 3) of the estimated parameter values reported in three independent studies (Tversky and Kahneman 1992; Camerer and Ho 1994; Wu and Gonzalez 1999).

In SBTCS, risk aversion is an option that can be switched on or off. When risk aversion is switched on, the actual audit rate is transformed using equation (5) into a perceived audit rate. This value is then used to determine when Defiant, Random, and Strategic taxpayer agents switch to full compliance. Defiant agents become fully complaint when $\hat{p} = 1/(1 + \phi)$. Random and Strategic agents become fully compliant when $\hat{p} \geq r$. When risk

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aversion is switched on, more taxpayers become fully compliant for audit probabilities up to the transition point in figure 3 (about 0.67) than when risk aversion is switched off.

Evolutionary Dynamics

On instantiation, the SBTCS model assigns (based on user-specified shares) behavioral archetypes to all individuals in a population of 10,000 taxpayer agents.[5] In each time period, taxpayers earn income (*ytrue*; described below) and decide how much of this income to report to the tax authority. The amount reported (*yreport*) is equal to $ytrue \times z$ using the instantiated value of z. Taxpayer net income (*ynet*) is determined as the difference ($ytrue - (taxRate \times yreport)$), where *taxRate* is a flat tax rate applied to reported income.

Net income may be reduced further, if the taxpayer is selected for an audit. If audited and no underreporting has occurred, the taxpayer is assumed not to incur any additional cost and net income is unchanged. If audited and underreporting is found (the auditor is presumed to discover all underreported income), the taxpayer remits all taxes due and an additional penalty based on taxes not reported. Net income is changed to reflect the audit outcome.

Once net income has been determined for all taxpayers, the model selects which agents survive to the next time period. The U.S. Small Business Administration (SBA 1991) finds that about 13 percent of small businesses cease operating each year. This attrition rate implies that the expected "life span" of a typical small business is five years. Thus, the SBTCS model assumes 1,300 firms are replaced in each time period by an equal number of new firms. Of these, 1,000 firms are assumed to cease operating due to bankruptcy and 300 firms exit for noneconomic reasons (e.g., retirement or career change). In both cases, exiting businesses are selected from the population at random. For firms experiencing bankruptcy, the following selection rule is used: If a randomly selected agent has net income less than the population median income, then they have a 90 percent chance of going bankrupt. However, if their net income is equal to or greater than the population median, there is a 10 percent chance of going bankrupt. This implies, other things being equal, that honest taxpayers are at greater risk for business failure due to their lower net income. The reporting compliance characteristics of the new firms are drawn using the same distributional parameters as the instantiated population unless the "neighborhood effects" option is switched on (described below).

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Return Selection Strategy

The SBTCS model provides several options for selecting tax returns for audit. One approach is random selection. This is the assumed selection method in the standard EU model of tax compliance and is also the method typically used in laboratory experiments. A second return selection method is a discriminant function (DIF) style audit. While the IRS uses tax return information and a statistical procedure based on Discriminant Analysis to identify which tax returns are likely candidates for an audit, the SBTCS model implements a DIF-like approach that selects returns in ascending order by reported income. For example, if the audit probability (p) is set to .05 and DIF-style audits are selected, then the 500 taxpayers (out of an initial population of 10,000) with the lowest reported incomes are selected for audit. A third option is to have a 50/50 mix of random and DIF-style audits. In this case if $p = .10$, then 500 taxpayer agents with the lowest reported incomes are audited and an additional 500 taxpayers are selected at random (excludes the first 500 audited returns selected using the DIF method). While stressing the point that the IRS DIF audit selection methodology is distinctly different than the approach described here, the goal of the two approaches is the same. For example, if only DIF-style audits are requested by the user, the SBTCS model will audit relatively fewer Honest taxpayer agents since these taxpayers, on average, report more income than any other archetype. The goal of IRS DIF methodology is the same—to identify the least compliant taxpayers for audit.

Neighborhood Effects

Several articles have recently investigated social network or neighborhood effects of taxpayer behavior (Fortin, Lacroix, and Villeval 2007; Korobow, Johnson, and Axtell 2007; Zaklan, Westerhoff, and Stauffer 2009). The SBTCS model includes an option to apply neighborhood effects when new firms replace expiring firms. When neighborhood effects are activated, a new agent is able to observe (or form an accurate opinion about) the compliance behavior of her four immediate neighbors (a Von Neumann neighborhood). If two or more neighbors are of type Honest and there are more Honest than Defiant neighbors, then she becomes Honest as well. If this condition is not met, then the new agent becomes Defiant if two or more neighbors are Defiant and there are more Defiant than Honest neighbors. If neither of these conditions is met, then the new agent is randomly assigned one of the four archetypes based on the initial distribution of archetypes in the population.

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Neighborhood effects provide an explicit mechanism for coordination to occur by allowing new agents to condition their behavior on what their neighbors are doing. However, for this mechanism to work, taxpayers must be able to observe or otherwise accurately assess the compliance behavior of other taxpayers in their immediate area. Performing model calibration both with and without neighborhood effects should help to determine whether such effects contribute to taxpayer reporting behavior.

Income Distribution

In each time period, true income (*ytrue*) is randomly allocated to all taxpayers, regardless of archetype. Income is drawn from one of two distributions specified by the user. An "experimental" distribution draws values from a uniform distribution with a minimum of 20 and a maximum of 100. This distribution is identical to that used to generate the experimental data in table 2 and figure 2A and B. A second distribution, designated "natural" in the model, draws income from a lognormal distribution with a median of 7 and mean of 16, which roughly replicates the distribution of incomes (in thousands of US dollars) from the weighted sample of taxpayers in the selected subsample of NRP data in table 1. Because Netlogo does not have a function to generate lognormal deviates, SBTCS generates these as follows:

$$x = e^{(2+1.25N)} \quad \text{where } N \sim \text{Normal}(\mu = 0, \quad \sigma = 1). \tag{6}$$

Miscellaneous

In addition to the above features, users specify the probability of audit, penalty rate, and number of simulations. By default, the random number seed is 123457, but users may tell the model to generate a unique random number seed for each simulation.

Model Calibration and Application

Model Calibration

Model calibration is performed by selecting an initial mix of behavioral archetypes that closely reproduces the mean and the bimodal distribution of compliance rates observed in laboratory experiments. Based on the analysis of subjects' reporting behavior,[6] the initial share of each behavioral archetype is set as follows: Honest = 15 percent, Defiant = 15 percent, Random = 20 percent, and Strategic = 50 percent. Other options selected are experimental

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Table 4. Calibration Results—Mean Reporting Rate

Audit Rate	Lab Experiment	No Risk Aversion	With Risk Aversion
0.00	0.288	0.286	0.286
0.05	0.404	0.327	0.421
0.10	0.475	0.376	0.489
0.30	0.558	0.527	0.679
0.40	0.672	0.607	0.738

Source: Author's calculations. Other assumptions: Initial share of Honest = Defiant = 15 percent, Random = 20 percent, Strategic = 50 percent, $T = 15$, $\Phi = 0.50$, Experimental income distribution.

income distribution, random audits, and ϕ (penalty rate) = 0.50. Simulations are run for fifteen time periods similar to the number of experimental rounds. Finally, because laboratory subjects were unable to observe others' behavior, neighborhood effects are switched off. Calibration results are shown in table 4.

By comparing the mean reporting compliance rates from the lab experiment shown in column 2 of table 4 (reproduced from table 2) with the simulated mean compliance rates in columns 3 and 4, we see that the simulation with risk aversion (overweighting of audit probabilities) outperforms the simulation with no risk aversion for audit rates of 0.10 or less. This would seem to indicate that individuals in tax compliance lab experiments overweight the risk of audit when rates are low. Historical audit rates in the United States reached a peak of approximately 0.05 in the early 1960s but have remained in the range of 0.01 in recent years (Bloomquist 2003a).

Figure 4 displays histograms of the distribution of individual reporting rates with risk aversion at the end of fifteen periods of simulation for each of the audit rates in table 4. These distributions closely resemble figure 2A and B with respect to their bimodal shape and the shifting of mass from 0 to 1 as the audit rate increases.

Application

In this section, we use the calibrated model to simulate changes in compliance behavior for a population of 10,000 small business owners using realistic values for audit and penalty rates. The following four scenarios are generated:

No risk aversion, no neighborhood effects
No risk aversion, neighborhood effects

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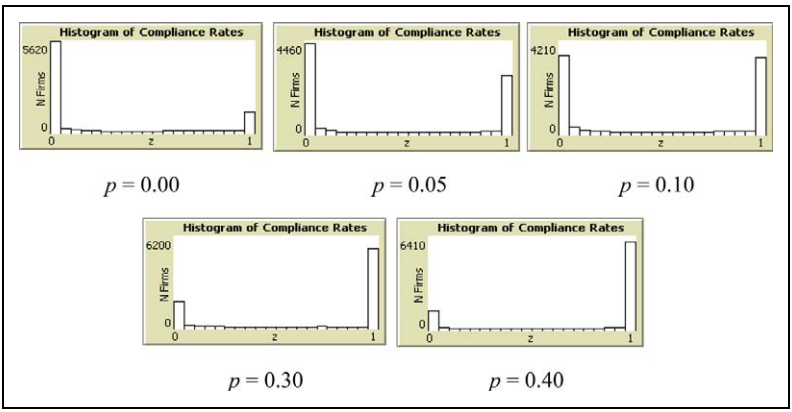


Figure 4. Histograms of simulation compliance rates by probability of audit.

Risk aversion, no neighborhood effects
Risk aversion, neighborhood effects

The output from each scenario is compared to the actual compliance of Schedule C filers in the NRP subsample to determine which combination of assumptions appears to have generated the observed behavior. Recall that the target mean reporting compliance rate (weighted) for this sample was 0.242 (table 1).

The following assumptions are made: audit probability (p) = .01, penalty (ϕ) = 0.50, an initial share of 50 percent for the Strategic archetype, 20 percent for the Random archetype, and 15 percent each for the Honest and Defiant archetypes, "natural" income distribution, the default random number seed and a 50/50 mix of random, and DIF-style audits for the return selection method. Finally, the number of time steps (T) equals 50. The mean reporting compliance rates for the four simulations are shown in table 5.

In all four simulations, the compliance rate produced by the model is higher than the target value of 0.242. This suggests that the model calibrated to behaviors observed in the lab may yield imprecise estimates. Therefore, following List and Levitt (2007), it is assumed that small business owners in the real world are self-selected to be less random in their reporting behavior than is likely true of college students in laboratory experiments. Consequently, the same four simulations were rerun assuming no Random agents, 70 percent Strategic agents and 15 percent Honest

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Table 5. Mean Reporting Compliance Rate Using Model Calibrated to Lab Experiments

Neighborhood Effects	No Risk Aversion	With Risk Aversion
No	0.326	0.363
Yes	0.414	0.495

Table 6. Mean Reporting Compliance Rate Using Model with No Random Agents

Neighborhood Effects	No Risk Aversion	With Risk Aversion
No	0.218	0.265
Yes	0.406	0.464

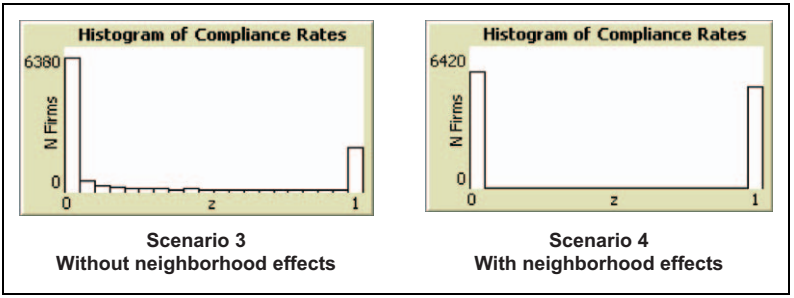


Figure 5. Histograms of compliance rates after fifty time periods.

and Defiant as before. All other parameters are unchanged. The results are shown in table 6.

From table 6, we see that the target value of 0.242 for mean reporting compliance rate is midway between the two simulations with no neighborhood effects. In contrast, the simulation values with neighborhood effects in table 6 are little changed from those in table 5.[7] These results are a further indication that the level of tax reporting compliance by small business owners in the real world does not necessarily result from taxpayers observing and mimicking others' reporting behavior but can result from a process of evolutionary adjustment over time to low levels of enforcement.

Figure 5 displays histograms for scenarios 3 (risk aversion and no neighborhood effects) and 4 (risk aversion with neighborhood effects). Comparing these two histograms to figure 1, it is clear that the simulation without

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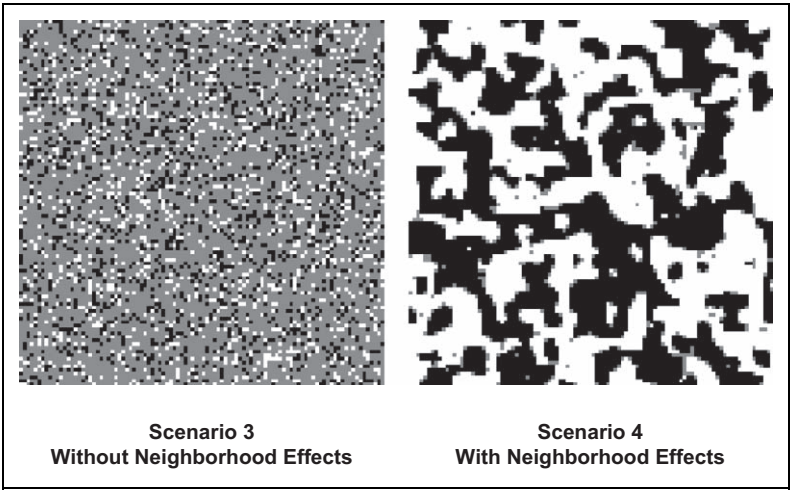


Figure 6. Agent landscapes after fifty time periods.

neighborhood effects (scenario 3) most closely resembles data from actual taxpayers. When neighborhood effects are introduced (scenario 4), the result after fifty time periods is that nearly all Strategic agents have been replaced by Honest and Defiant agents.

The impact of introducing neighborhood effects is depicted in figure 6. In these landscapes, Honest agents are coded in black, Defiant agents in white, and Strategic agents are coded in grey. The landscape for scenario 3 clearly shows a majority (approximately 70 percent) of agents are of the Strategic archetype with roughly even shares of Honest and Defiant agents randomly scattered about. Scenario 4, in contrast, shows nearly all Strategic agents have been replaced by Honest and Defiant agents. However, the situation at $T = 50$ does not represent a condition of long-run stability. If the simulation is allowed to run for a very long time, eventually the landscape will become completely populated with Defiant Agents.[8] This outcome results from the assumption that firms that evade more (and have more after tax net income) are less likely to go bankrupt.

Conclusion

This article has presented an agent-based model of small business taxpayer reporting compliance based on evolutionary game theory. The model is

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calibrated to observations of human subjects' reporting behavior in tax compliance laboratory experiments. The calibrated model is used to generate alternative scenarios to determine the most plausible combination of assumptions that reproduces the behavior of taxpayers observed in a random audit study. Based on the calibration and simulation results, it appears that both subjects in lab experiments and actual small business taxpayers are risk averse (i.e., overweight the probability of an audit) and taxpayers in the real world do not base their reporting compliance decisions on their neighbors' behavior.

An extension of this model would be to allow for the costs of compliance including burden costs, the cost of hiring a tax professional, and so on. Another enhancement would be to incorporate a broader range of taxpayer behaviors, such as the "bomb crater" response mentioned previously. Finally, the firms portrayed in this model are limited to relatively simple privately owned sole proprietorships. A future extension would be to include other types of firms, such as partnerships and corporations as well as firms varying in number of employees or assets.

Finally, this article offers a case study showing how data from laboratory experiments and the techniques of agent-based modeling can be used in combination to investigate real-world phenomena (Duffy 2006). In the current case, lab experiments were used to provide insights into taxpayer behavior that are unobservable from either field surveys or random audit studies. Such behaviors include learning and responses to changing enforcement parameters. While the model calibrated from experimental data could not, at first, replicate the compliance behavior of audited taxpayers, the model output provided insights into appropriate adjustments to make (i.e., eliminating all Random archetype agents and increasing the initial number of Strategic agents). This illustrates how laboratory experiments and agent-based modeling can both complement and reinforce each other while also indicating areas where more research is needed.

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Notes

1. A risk-neutral taxpayer should underreport 100 percent of tax due, whereas a risk-averse taxpayer would underreport some positive amount of tax but less than 100 percent.
2. Allingham and Sandmo (1972) also mention this extension in the context of a loss of reputation on the part of the taxpayer.
3. Alm and McKee (2004) suggest this is conceptually similar to the IRS' Discriminant Function (DIF) formula used to select returns for operational audits.
4. The version of the tax compliance model in (1) is the variant proposed by Yitzhaki (1973) in which a fine is imposed on the amount of tax evaded, unlike in the original Allingham and Sandmo (1972) model that assumes the fine is applied to the amount of income evaded.
5. The modeling environment is NetLogo (version 4.0.4; Wilensky 1999).
6. See the discussion on pages 10–12 above.
7. Results from 100 simulations using different random number seeds yielded results similar to those in table 6.
8. At time $T = 5{,}000$, there are 9,272 Defiant, 687 Honest, and 41 Strategic agents.

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Bio

Kim Bloomquist is a senior economist in the Internal Revenue Service's Office of Research in Washington, D.C. In that capacity, his research broadly addresses various aspects of tax noncompliance with specific emphasis on agent-based models of taxpayer behavior and the design and implementation of human subject tax compliance laboratory experiments.

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